

0.1 - Scientific Attitude, Critical Thinking, Developing Arguments

What is Scientific Attitude?

- It is embracing curiosity, skepticism, and humility to use Critical thinking
- We use evidence and and Critical thinking to examine assumptions, assess the source, discern hidden bias, evaluate evidence, and assess conclusions
- Directed, problem-focused thinking for solutions/errors. Essential to examining how valid a hypothesis is in meaning of research
- Scientific inquiry can help sift reality from illusion

Scientific Attitude Components

Curiosity

- Does it work? Can it be tested and confirmed?

Skepticism

- Is it true? How do you know? Cannot believe everything, but also cannot write off everything

Humility

- How can I explore deeper? Willingness to follow new ideas as not everything is as predicted. Be surprised and open to beliefs and original perceptions being different.

Critical Thinking

- Reevaluating and interpreting information deeper to recognize and eliminate bias, better evaluate evidence, question and assess sources, and come to conclusions.
- Science based answers are better than “common sense” based answers because of human tendency to have biases.

Research & Thinking Biases

Cultural norms

- Expectations and rules shared among a group of people that guides accepted behaviors

Cognitive biases

- Systematic patterns of thinking errors that affect judgment and decision-making.

Confirmation bias

- Tendency to search for or interpret information that confirms one's beliefs.

Hindsight bias

- Belief that one “knew it all along” after learning an outcome.

Overconfidence

- Overestimating the accuracy of one's beliefs or judgments.

0.2 - Need for Psychological Science

Psychology

- The study of the mind and behavior through the use of research and experimental methods to understand thoughts, emotions, and behaviors
- Base of psychology is observation

Road Blocks to Critical Thinking

Hindsight Bias

- After learning an outcome, individuals have a tendency to believe they would have foreseen it—“I knew it all along.” This bias can lead people to criticize poor, risky decisions while praising successful ones, even when both involve similar uncertainty.

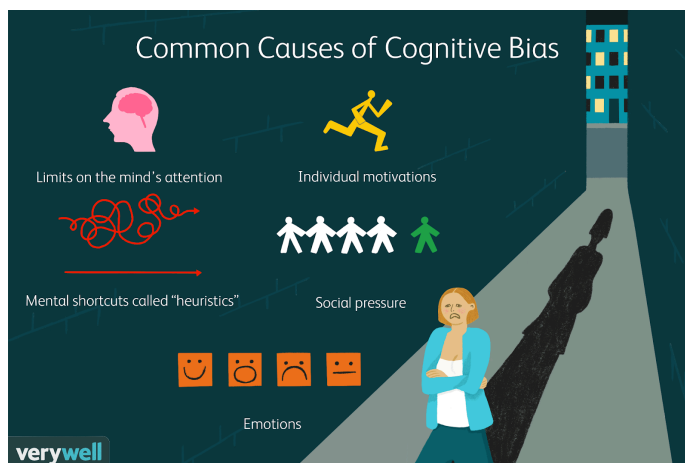
When findings align with “common sense,” individuals may assume they already understood the concept and fail to question or analyze it more deeply.

Overconfidence

- Overconfidence often results from a tendency to seek out information that confirms existing judgments. This inflated sense of certainty can drive quicker thinking, but not necessarily more accurate thinking. As a result, individuals may become blind to errors or alternative perspectives, even when their answers are incorrect.

Perceiving Patterns in Random Events

- Many people find a random and unpredictable world unsettling. As a result, they tend to perceive patterns or order in random events, as doing so provides a sense of control and reduces stress.

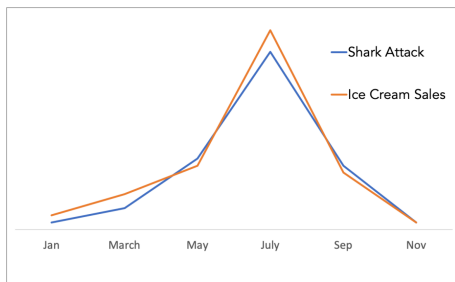


Confirmation Bias

- The tendency to seek out, interpret, and remember information that confirms our pre-existing notions and beliefs while also ignoring and discounting contradictory evidence

Key Idea

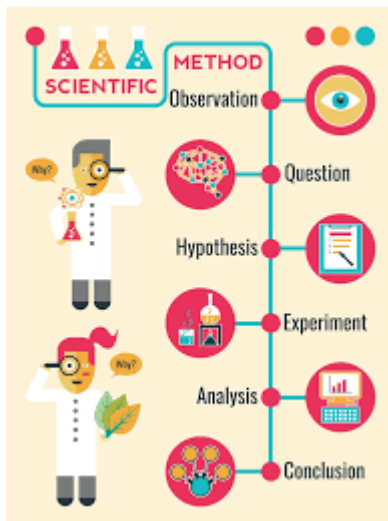
- Correlation is NOT Causation



How is psychology a science?

- Psychology uses scientific methods such as observation, experimentation, and testing to draw conclusions and make educated guesses.

0.3 - The Scientific Method



Core Concepts

- **Hypothesis:** testable explanation or prediction for a specific phenomenon
- **Falsifiability:** Your hypothesis can be proven true or false

- **Operational Definition:** Specifies how a concept or term is going to be defined or measured in practice in clear instructions
- **Null Hypothesis (H_0)** States that there is no relationship or difference between variables.
 - Default stance when doing statistical testing
 - e.g. "There is no significant relationship between caffeine consumption and academic performance."
- **Alternative Hypothesis (H_a):** Predicts that a relationship or effect does exist between the variables
 - Tested against the null hypothesis

Reliability & Validity

Reliability

- Consistently getting the same result when doing something multiple times.
→ If a study is reliable, it should give similar results every time it is repeated.
- Replication is confirmation

Validity

- Testing how true or accurate something actually is.
→ A study can be reliable but not valid (consistent but wrong).

Generalizability & Sampling Basics

Generalizability

- How well the results from a sample can be applied to the larger population.
→ If your sample is good, you can confidently generalize your results.

Population

- The entire group of people you want to study.
→ Example: all students in the U.S., all humans, all students at Darnell Cookman.

Sample

- A smaller group taken from the population.
→ Example: one class, one school, one city.

Sampling

- The process of choosing people from the population to be in your sample.

Sampling Methods & Issues

Sampling Bias

- When the sample is chosen in a way that does NOT represent the population.
→ Happens when researchers pick people based on what they want instead of accuracy.
→ Example: only surveying athletes about school stress.

Convenience Sampling

- Choosing people who are easiest or closest to you.
→ Example: surveying your friends or your own class.

Representative Sampling

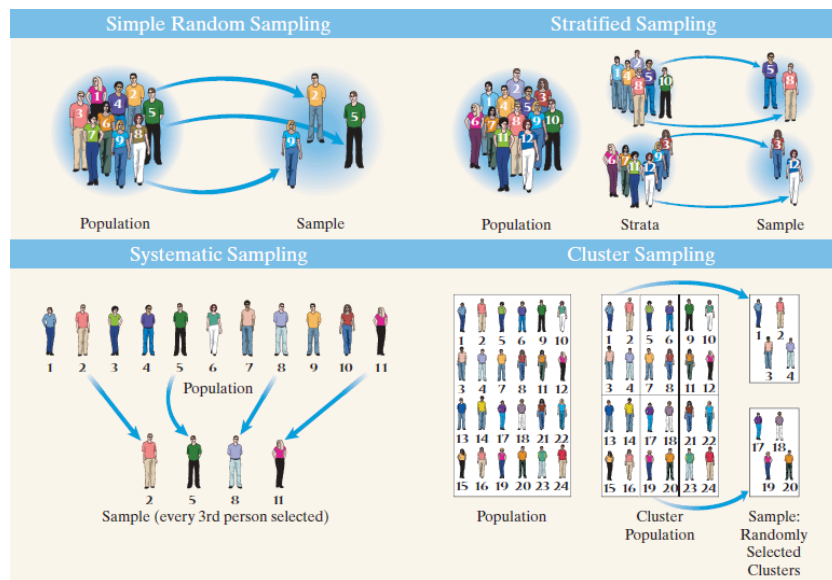
- A sample that matches the population's key characteristics (age, gender, race, etc.).
→ This makes your results more accurate.

Random Sampling

- Everyone in the population has an equal chance of being chosen.
→ No one is specifically picked.

Why is representative sampling important?

- It helps ensure your data actually reflects the population, making your conclusions more accurate and trustworthy.



Types of Data

Qualitative Data

- Non-numerical, descriptive data.
→ Example: opinions, feelings, descriptions.

Quantitative Data

- Numerical data that can be measured.
→ Example: scores, percentages, statistics.

Surveys & Bias

Pros of Surveys

- Quick, efficient, and can collect a lot of data from many people.

Cons of Surveys

- You can't always tell if people are telling the truth or taking it seriously.

Self-Report Bias

- When people give inaccurate answers about themselves.
→ Often to look better or avoid embarrassment.

Social Desirability Bias

- When people answer in a way they think is more acceptable or “correct.”
→ Example: saying you study more than you actually do.

Scientific Process & Validation

Peer Review

- Other experts in the same field examine your study before it is accepted.
→ Helps check for mistakes and improve accuracy.

Replication

- When another researcher repeats your study and gets the same results.

Why is replication important?

- If the same results happen again, it proves the findings are reliable and not just by chance.

Theories

- Explanations that apply an integrated set of principles → used to organize observations and generate hypotheses
- Theories can bias observations → we may see what we expect or already believe

A useful theory:

- Organizes observations clearly
- Implies predictions that can be tested
- Strong theories stimulate further research → may be revised to better organize and predict behavior
- Replication = confirmation of findings across repeated studies

Research Methods

Experimental Methodologies

- Designed to test a hypothesis to establish a cause and effect relationship between a dependent and independent variable using a controlled scientific environment. Requires random assignment.

Non Experimental Methodologies

- Used for research when controlled experimenting would be impossible or unethical. Just meant to describe behavior and cannot establish cause and effect relationship between variables.

Case study

- Focused examination of a specific person, group, event, or situation for unique detailed insight on a topic.

Correlational study

- Determines the relationship and strength between two variables.

Meta-analysis

- Combining results across many studies to make a conclusion.

Naturalistic Observation

- Gathering data by observing individuals in natural environments and a real world setting.

0.4 - Correlation and Experimentation

Correlation

- Correlation: Relationship between two variables.

Correlation coefficient (r)

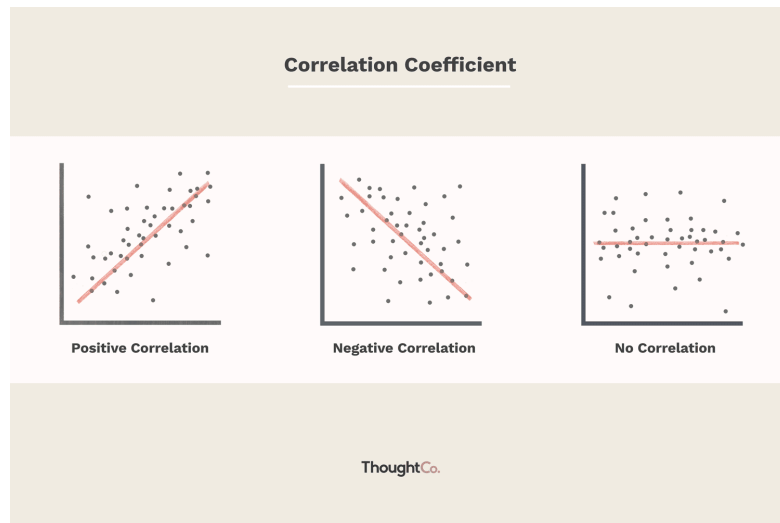
- Strength/direction of correlation (-1 to +1).

Positive correlation

- Both variables increase together.

Negative correlation

- One increases while the other decreases.



Data & Statistics

Mean

- Average score
 - Sum of all values divided by number of values.
 - Can be skewed/affected by extreme values (outliers) → can pull the mean up or down

Median

- Middle score, separating lower half and higher half

Mode

- Most frequent score
 - Value appearing most often in a dataset.
 - A dataset can have one mode, more than one mode, or no mode at all

Range

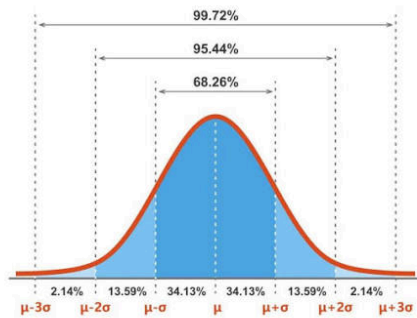
- Difference between highest and lowest

Standard deviation

- Measure of data spread around the mean
 - Indicates how much scores vary from the average
 - Low standard deviation = scores close to the mean (clustered)
 - High standard deviation = scores spread out over a wider range

Normal curve/Bell Curve

- Symmetrical bell-shaped distribution
 - Most scores cluster around the mean, with fewer scores at the extremes
 - Mean, median, and mode are all equal at the center.
 - 68–95–99.7 rule:
 - 68% of data within 1 standard deviation
 - 95% within 2 SD
 - 99.7% within 3 SD



Percentile rank

- Percentage of scores below a given score
 - 90th percentile = scored higher than 90% of people

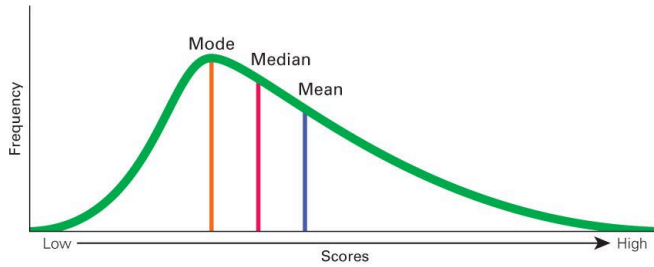
Positive skew

- Tail extends right; mean > median
 - Most scores are low, with a few very high outliers pulling the mean up
 - E.g. income distribution (few very high incomes)

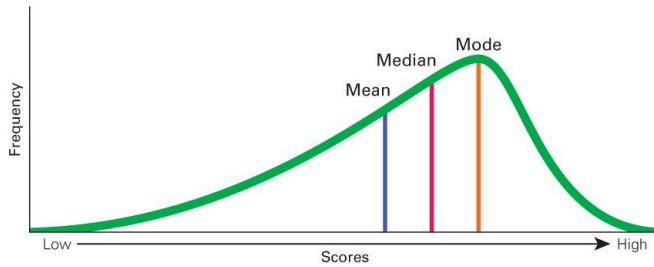
Negative skew

- Tail extends left; mean < median
 - Most scores are high, with a few very low outliers pulling the mean down

- easy test where most students score high



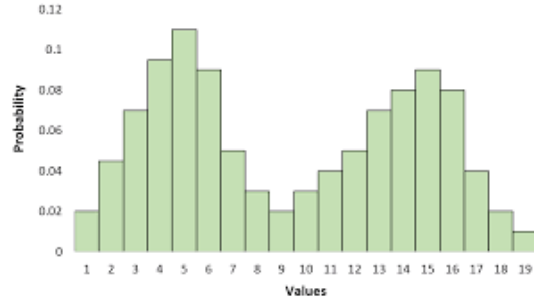
(a) Right-skewed distribution



(b) Left-skewed distribution

Bimodal distribution

- Two peaks in data with a dip towards the middle



Data Types

Descriptive data

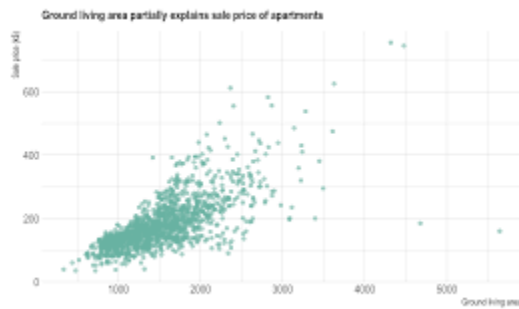
- Summarizes data (mean, graphs).

Inferential data

- Draws conclusions about a population.

Scatterplot

- Graph showing relationship between variables



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Effect size

- Strength of a relationship or difference.

Statistical significance

- Likelihood results are not due to chance ($p < .05$)
 - Means there is less than a 5% probability that the results happened by chance
 - Used to determine if results are meaningful and can be trusted

Experimental Design

Independent variable (IV)

- Factor manipulated by the researcher.

Dependent variable (DV)

- Outcome measured in the experiment.

Variables

- Any factors that can change in a study.

Confounding variable

- Outside factors that affect the results.

Third variable problem

- A hidden variable influences both variables in a correlation.

Experimental group

- Group exposed to the IV.

Control group

- Group not exposed to the IV.

Random assignment

- Randomly placing participants into groups.

Placebo

- Inactive substance or condition.

Placebo effect

- Improvement due to expectations, not treatment.

Single-blind procedure

- Participants are unaware of whether they are assigned to the treatment or control (placebo) group; however, researchers know.

Double-blind procedure

- Neither the participants nor the researchers interacting with them know who is receiving a particular treatment or placebo

Experimenter bias

- Researcher's expectations influence results
 - Researcher might subconsciously interpret data to support their preconceived hypothesis rather than objectively recording the findings

Bias in Research

- **Wording Effects**
 - Question phrasing can influence responses

- Researchers aim to reduce **social desirability bias** (tendency to give “acceptable” answers)
- Often paired with behavioral measures to reduce **self-report bias** (inaccurate responses)

Wording effect

- The way a question is phrased can influence responses.
- Researchers aim to reduce **social desirability bias**
- Often paired with behavioral measures to reduce **self-report bias**

Self-report bias

- Inaccurate responses in surveys

Social desirability bias

- Responding in a way viewed favorably by others; a tendency to give “acceptable answers”

Ethics

Peer review

- Evaluation by other experts before publication.

Institutional Review Board (IRB)

- Committee ensuring ethical standards.

Informed consent

- Participants are fully informed before agreeing.

Informed assent

- Minors or those unable to physically consent themselves agree through parental consent

Protection from harm

- Participants must not be harmed in any way, physically or mentally

- Participants have the **Right to Withdraw**

Confidentiality

- Keeping participant information private and secure

Deception

- Misleading participants from the true purpose, procedures, or conditions of a study (must be justified and ethical).

Debriefing

- Conversation after the experiment is over where researchers explain the true purpose of the study to the participants, reveal any deception used, and address misconceptions

Confederates

- Actors working alongside researchers in a study, they are not participants.

Survey Tools

Structured interviews

- Standardized questions asked to all participants.

Likert scales

- Rating scale (e.g., strongly agree → strongly disagree)
-